

De Novo Binder Docking Album

Top 5 designed miniprotein binders — Aβ42 (Alzheimer's) target

BindCraft campaign • ranked by Stage-5 composite score • target 9CO4 (receptor-bound Aβ42 fibril)



designed binder



Aβ42 fibril target (9CO4)

Each panel is an AlphaFold2-backpropagation binder docked onto the Aβ42 fibril target — cartoon with a translucent molecular-surface glow, rendered in the campaign house style. All five bind laterally across the fibril, engaging the designed hotspot cluster (Aβ42 10–16).

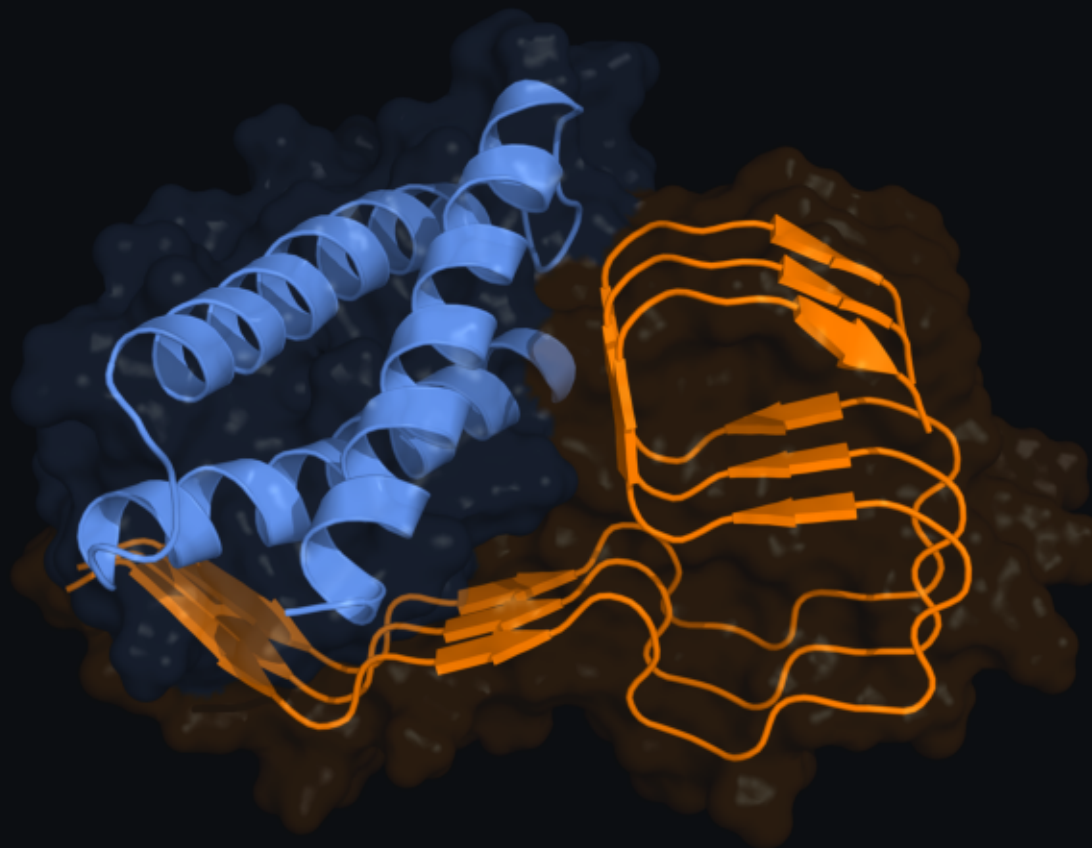
#	Design (scaffold_mpnn)	Len	i-pTM	i-pAE	dG	SC	Score	Iface res (bind/tgt)
1	l82_s967366_mpnn11	82	0.79	0.17	-71.8	0.76	0.793	16/17
2	l89_s578974_mpnn11	89	0.80	0.19	-70.8	0.68	0.733	18/23
3	l90_s311742_mpnn16	90	0.86	0.15	-88.8	0.76	0.713	23/22
4	l71_s843399_mpnn18	71	0.80	0.18	-76.6	0.71	0.708	20/16
5	l90_s311742_mpnn3	90	0.86	0.15	-88.0	0.76	0.696	21/21

Chain A = Aβ42 target (9CO4, C/E/G fibril chains) • Chain B = designed binder

Prepared 2026-07-05 • ab42-binder-design campaign

#1 l82_s967366_mpnn11

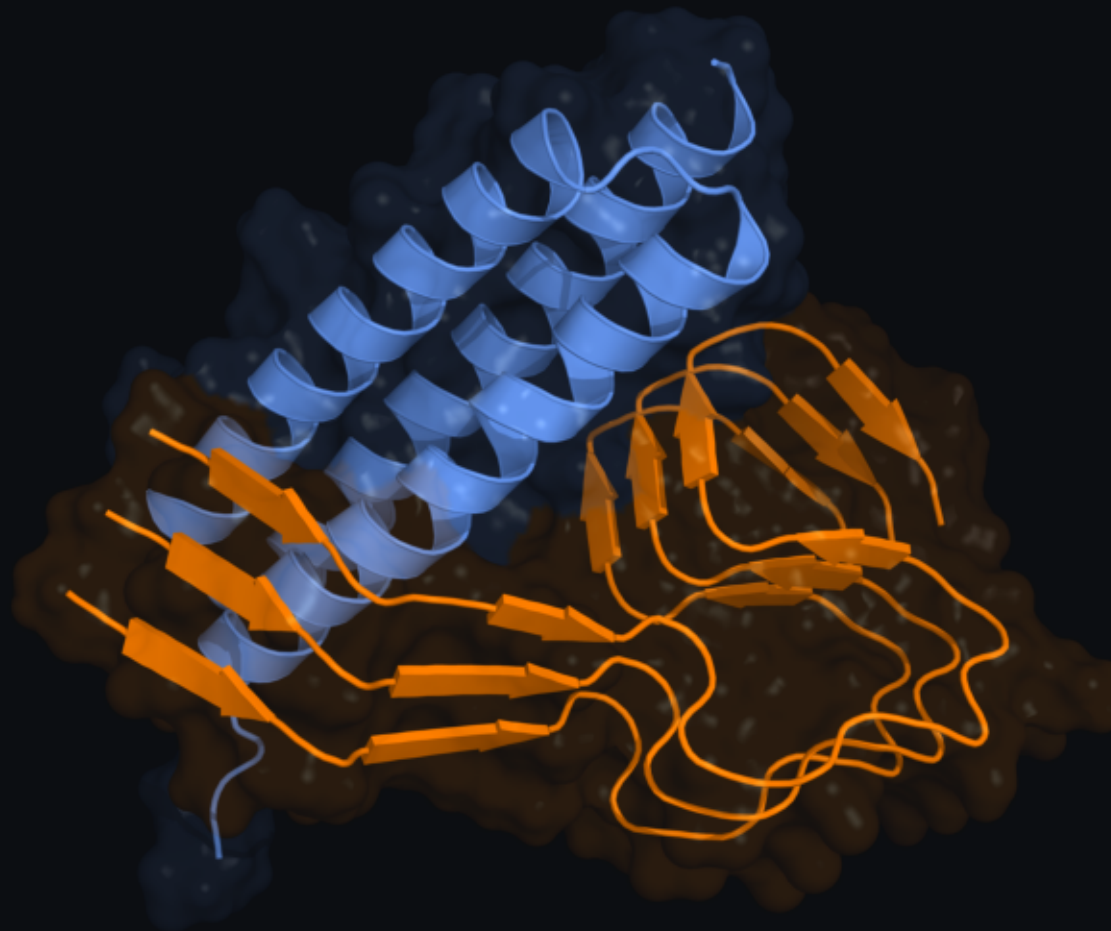
scaffold s967366 • 82 aa • i-pTM 0.79 • i-pAE 0.17 • ΔG -71.8 kcal/mol • shape-comp 0.76 • score 0.793



Binder (blue) docked laterally across the A β 42 fibril (orange), burying 16 binder residues against 17 target residues and engaging all 6 designed hotspots (A β 42 10,11,13-16). i-pTM 0.79, interface PAE 0.17 Å, Rosetta ΔG -71.8 kcal/mol.

#2 l89_s578974_mpnn11

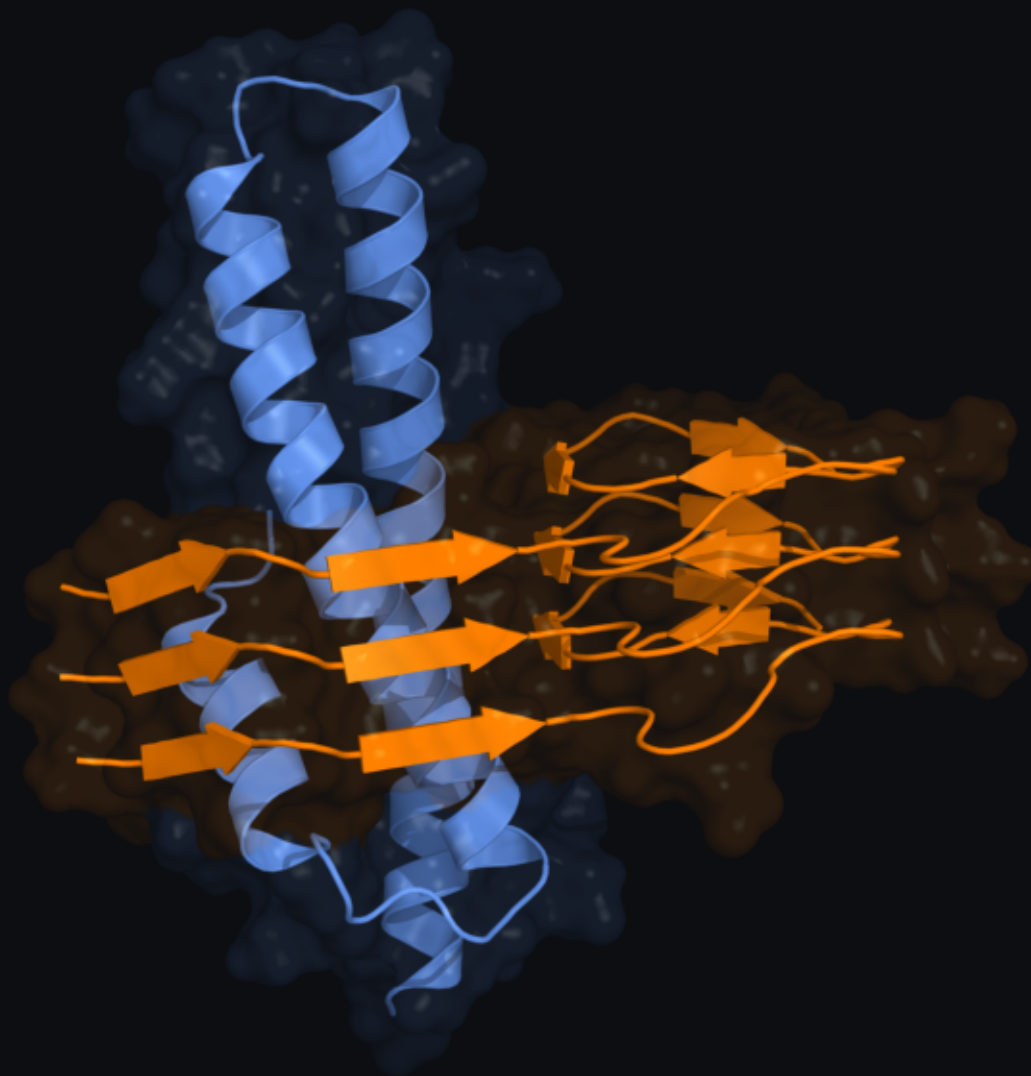
scaffold s578974 • 89 aa • i-pTM 0.80 • i-pAE 0.19 • ΔG -70.8 kcal/mol • shape-comp 0.68 • score 0.733



Binder (blue) docked laterally across the A β 42 fibril (orange), burying 18 binder residues against 23 target residues and engaging all 6 designed hotspots (A β 42 10,11,13-16). i-pTM 0.80, interface PAE 0.19 Å, Rosetta ΔG -70.8 kcal/mol.

#3 I90_s311742_mpnn16

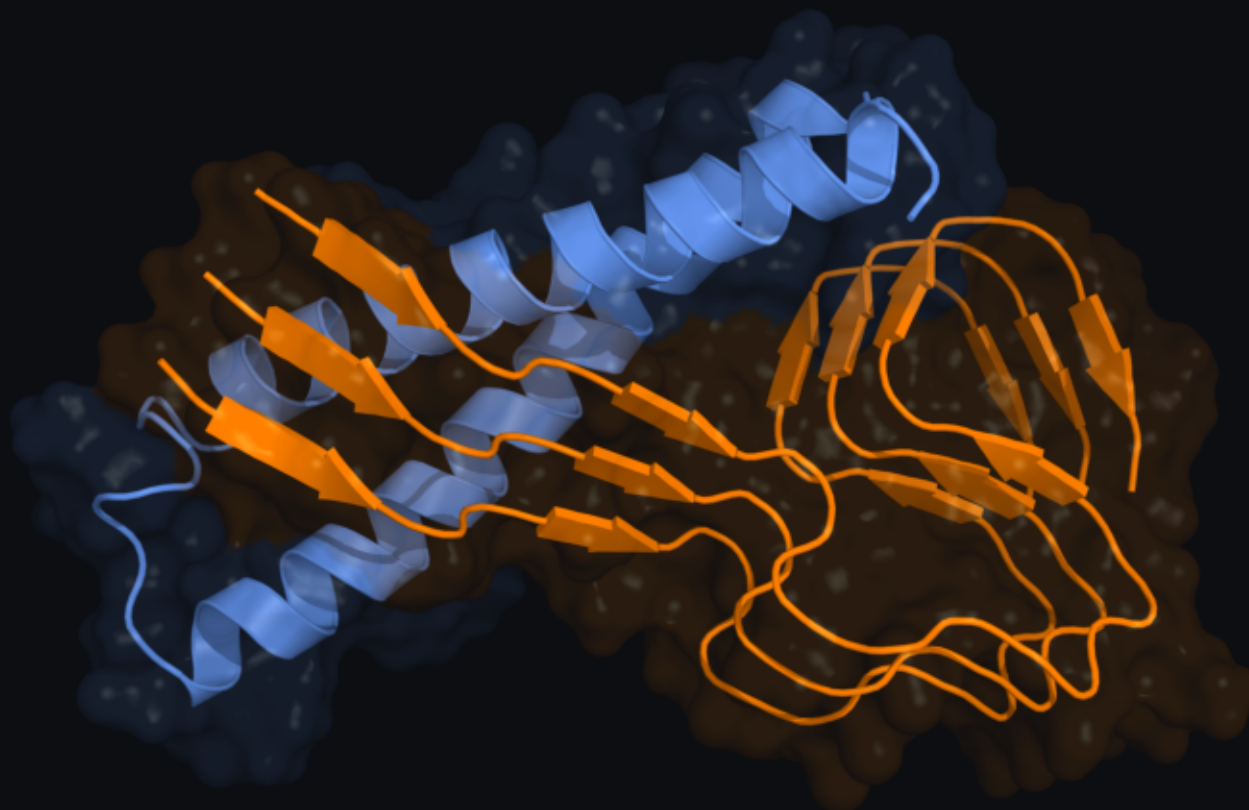
scaffold s311742 • 90 aa • i-pTM 0.86 • i-pAE 0.15 • ΔG -88.8 kcal/mol • shape-comp 0.76 • score 0.713



Binder (blue) docked laterally across the A β 42 fibril (orange), burying 23 binder residues against 22 target residues and engaging all 6 designed hotspots (A β 42 10,11,13-16). i-pTM 0.86, interface PAE 0.15 Å, Rosetta ΔG -88.8 kcal/mol.

#4 l71_s843399_mpnn18

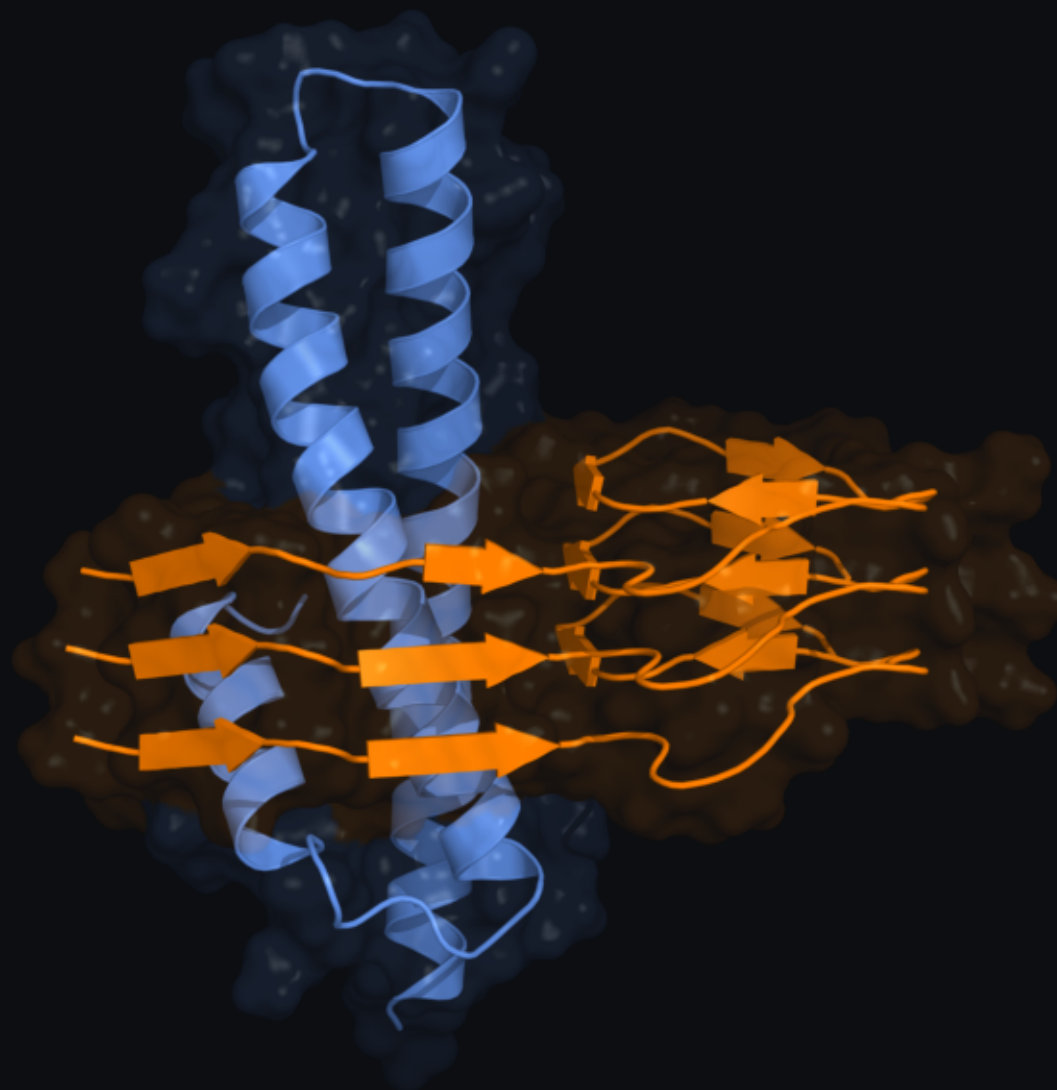
scaffold s843399 • 71 aa • i-pTM 0.80 • i-pAE 0.18 • ΔG -76.6 kcal/mol • shape-comp 0.71 • score 0.708



Binder (blue) docked laterally across the A β 42 fibril (orange), burying 20 binder residues against 16 target residues and engaging all 6 designed hotspots (A β 42 10,11,13-16). i-pTM 0.80, interface PAE 0.18 Å, Rosetta ΔG -76.6 kcal/mol.

#5 I90_s311742_mpnn3

scaffold s311742 • 90 aa • i-pTM 0.86 • i-pAE 0.15 • ΔG -88.0 kcal/mol • shape-comp 0.76 • score 0.696



Binder (blue) docked laterally across the Aβ42 fibril (orange), burying 21 binder residues against 21 target residues and engaging all 6 designed hotspots (Aβ42 10,11,13-16). i-pTM 0.86, interface PAE 0.15 Å, Rosetta ΔG -88.0 kcal/mol.